

PRACTICAL-2

```
1 • CREATE DATABASE bookflow_df;
2 • USE bookflow_df;
3 • CREATE TABLE Books (
4     book_id INT PRIMARY KEY,
5     title VARCHAR(100) NOT NULL,
6     isbn VARCHAR(20) UNIQUE,
7     published_year INT CHECK (published_year < 2027)
8 );
9 • INSERT INTO Books (book_id, title, isbn, published_year) VALUES
10 (1, 'The Great Gatsby', '9780743273565', 1925),
11 (2, 'To Kill a Mockingbird', '9780061120084', 1960),
12 (3, '1984', '9780451524935', 1949);
13 • SELECT * FROM Books;
14 • CREATE TABLE Members (
15     member_id INT PRIMARY KEY,
16     full_name VARCHAR(100),
17     email VARCHAR(100) UNIQUE
18 );
19 • INSERT INTO Members (member_id, full_name, email) VALUES
20 (101, 'John Smith', 'john.smith@email.com'),
21 (102, 'Emma Wilson', 'emma.wilson@email.com'),
22 (103, 'Michael Brown', 'michael.brown@email.com');
23 • SELECT * FROM Members;
24 • CREATE TABLE Loans (
25     loan_id INT PRIMARY KEY,
26     member_id INT,
```

```

25     loan_id INT PRIMARY KEY,
26     member_id INT,
27     book_id INT,
28     loan_date DATE,
29     FOREIGN KEY (member_id) REFERENCES Members(member_id),
30     FOREIGN KEY (book_id) REFERENCES Books(book_id)
31 )
32 ✖ INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES
33 (1, 101, 1, '2025-01-05'), (2, 102, 2, '2025-01-08'),
34 (3, 103, 3, '2025-01-10'), (4, 101, 2, '2025-02-01'),
35 (5, 102, 1, '2025-02-05'), (6, 103, 2, '2025-02-12'),
36 (7, 101, 3, '2025-03-01'), (8, 102, 3, '2025-03-07'),
37 (9, 103, 1, '2025-03-15'), (10, 101, 1, '2025-04-01');
38 • SELECT * FROM Loans;
39 • SELECT m.full_name AS Member_Name, b.title AS Book_Title
40 FROM Loans l
41 INNER JOIN Members m ON l.member_id = m.member_id
42 INNER JOIN Books b ON l.book_id = b.book_id;
43 • SELECT published_year, COUNT(book_id) AS Total_Books
44 FROM Books
45 GROUP BY published_year
46 ORDER BY published_year;
47 • CREATE TABLE Donation_History (
48     donation_id INT PRIMARY KEY,
49     book_id INT,
50     donor_name VARCHAR(100),

```

```

35 (5, 102, 1, '2025-02-05'), (6, 103, 2, '2025-02-12'),
36 (7, 101, 3, '2025-03-01'), (8, 102, 3, '2025-03-07'),
37 (9, 103, 1, '2025-03-15'), (10, 101, 1, '2025-04-01');
38 SELECT * FROM Loans;
39 SELECT m.full_name AS Member_Name, b.title AS Book_Title
40 FROM Loans l
41 INNER JOIN Members m ON l.member_id = m.member_id
42 INNER JOIN Books b ON l.book_id = b.book_id;
43 SELECT published_year, COUNT(book_id) AS Total_Books
44 FROM Books
45 GROUP BY published_year
46 ORDER BY published_year;
47 • CREATE TABLE Donation_History (
48     donation_id INT PRIMARY KEY,
49     book_id INT,
50     donor_name VARCHAR(100),
51     donation_date DATE,
52     FOREIGN KEY (book_id) REFERENCES Books(book_id)
53 );
54 START TRANSACTION;
55 INSERT INTO Books (book_id, title, isbn, published_year)
56 VALUES (4, 'Animal Farm', '9780451526342', 1945);
57 INSERT INTO Donation_History (donation_id, book_id, donor_name, donation_date)
58 VALUES (1, 4, 'Raj Kumar', CURDATE());
59 COMMIT;

```